

Case 1

02582 Computational Data Analysis

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Introduction/Data description

The task is to predict a continuous response y from a 100-dimensional feature matrix X given only $n = 100$ labelled observations — a canonical high-dimensional regime ($n = p$) in which standard OLS is undefined and regularisation is mandatory. The Expected Prediction Error decomposes as $\text{EPE} = \sigma_\epsilon^2 + \text{Bias}^2 + \text{Var}$, and navigating this bias-variance tradeoff is the central challenge.

The training file `case1Data.csv` contains $n = 100$ rows and 101 columns: one response y and $p = 100$ predictors. Of these, 95 are continuous (`x_01–x_95`) and 5 are ordinal categorical variables (`C_01–C_05`) taking integer values in $\{71, 72, 73, 74, 75\}$. The response y ranges from 39.7 to 375.8 with mean 211.5 and standard deviation 72.5. A correlation analysis reveals a sparse signal: the leading predictor `x_32` achieves $|r| = 0.77$ with y , followed by `x_36` ($|r| = 0.48$) and `x_31` ($|r| = 0.43$), while most other features show weak associations — suggesting that only a handful of predictors carry genuine signal. The training set contains 1,489 missing entries ($\approx 15\%$ per column), and the prediction set `case1Data_Xnew.csv` ($n_{\text{new}} = 1,000$ observations) contains 15,061 missing entries, making imputation a critical design choice.

Model and method

All preprocessing and modelling steps are encapsulated in `scikit-learn Pipeline` objects so that every transformation is fitted exclusively on training-fold data, preventing any data leakage. The full pipeline is: **(1)** impute missing values, **(2)** standardise continuous features (zero mean, unit variance), **(3)** encode categorical variables, **(4)** fit a penalised linear model.

We compared eight model families using 5-fold cross-validated RMSE:

Model	CV RMSE	Notes
ElasticNet (CV-tuned)	23.2	L1 + L2, best balance
LASSO (CV-tuned)	24.1	Pure L1, ultra-sparse
Ridge (CV-tuned)	25.3	L2 only
PCR ($d = 50$ components)	31.8	PCA then OLS
PLS ($d = 15$ components)	32.6	Maximises $\text{Cov}(X\beta, y)$
Gradient Boosting	38.1	High variance at $n = 100$
Random Forest	48.3	Insufficient data for splits
SVR (RBF kernel)	50.1	Kernel trick fails at $n = 100$

Linear penalised models consistently dominate: with $n = p = 100$, tree-based and kernel methods have too few observations to reliably estimate non-linear interactions. PLS and PCR, despite being designed for $p \approx n$, compress X into latent components without zeroing out noise predictors, whereas ElasticNet directly shrinks individual coefficients to zero — a more surgical regularisation when the signal is sparse. ElasticNet minimises

$$\frac{1}{2n} \|y - X\beta\|_2^2 + \alpha \rho \|\beta\|_1 + \frac{\alpha(1 - \rho)}{2} \|\beta\|_2^2,$$

with α and ρ tuned by inner 3-fold CV.

Beyond ElasticNet, we explored two additional sparse methods. **Adaptive LASSO** (Zou 2006) assigns predictor-specific weights $w_j = 1/|\hat{\beta}_j^{\text{Ridge}}|^\gamma$, penalising weak predictors more aggressively than strong ones. Unlike the standard LASSO, it satisfies the *oracle property* (consistent support recovery). With $\gamma = 1.0$ (chosen by grid search over $\{0.25, 0.5, \dots, 3.0\}$), CV RMSE improved to 22.07. **Stability Selection** (Meinshausen & Bühlmann 2010) runs Adaptive LASSO on $B = 100$ sub-samples of size $n/2$, retains only features selected in more than 50% of iterations (≈ 8 stable features), then refits OLS on this support. This reduced the inter-fold standard deviation from 4.69 to 3.92.

Model selection

Hyperparameter selection is performed by nested cross-validation: for each outer fold (5-fold), an inner 3-fold CV loop selects α , ρ , and γ using only the training observations of that fold. The outer fold score is computed on the held-out observations without any further fitting. This strict nesting prevents optimistic bias in the model comparison above.

For the final ensemble, optimal weights $(w_1, w_2) = (0.60, 0.40)$ for Adaptive LASSO and Stability OLS were selected by exhaustive grid search over the out-of-fold predictions of the 5-fold CV run, without re-running any fold.

Missing values

We benchmarked three strategies, all evaluated via 5-fold CV RMSE:

Strategy	CV RMSE	Gain vs. median
Median imputation	33.2	—
KNN imputation ($k = 5$)	26.2	−21%
MICE ($t = 2$)	23.2	−31%

MICE (Multiple Imputation by Chained Equations) was implemented via `IterativeImputer` with a `BayesianRidge` regressor and $t = 2$ iterations. At each iteration, every feature with missing values is regressed on all others and missing entries are replaced by predicted values. Median imputation ignores all inter-feature correlations; KNN captures local structure; MICE models the full joint distribution. The 31% RMSE reduction confirms that imputation quality is the single most impactful modelling decision in this dataset.

Factor handling

Column `C_02` has zero variance (constant value 72) and is dropped before any modelling. The remaining four categorical columns (`C_01`, `C_03–C_05`) take ordered integer values in $\{71, \dots, 75\}$. Although the levels are ordered, we chose *one-hot encoding* rather than ordinal encoding: the spacing between levels is unknown and need not be uniform, so treating the codes as numeric imposes a potentially false metric. With 5 levels per feature, one-hot encoding adds only $4 \times 4 = 16$ binary columns, bringing the total dimensionality to 115 after encoding. Missing categorical values are imputed by mode before encoding. The option `handle_unknown='ignore'` silently zeros out any unseen level in the test set.

Model validation

We used **5-fold cross-validation** (fixed `random_state=42`) throughout. At each fold: **(i)** MICE is fitted on the 80 training observations only; **(ii)** scaling parameters and encoder are fitted on the same 80 observations; **(iii)** hyperparameters are tuned via an inner 3-fold loop on those 80 observations;

(iv) predictions are generated for the 20 held-out observations using the pipeline fitted in steps (i)–(iii). No information from held-out observations enters any fitting step, guaranteeing an unbiased EPE estimate.

We chose 5 folds over 10 because MICE is computationally expensive and each 5-fold held-out set still contains 20 observations, sufficient for a reliable fold-level RMSE. The 5-fold CV estimate carries a slight upward bias (each fold trains on 80 rather than 100 observations), so our estimate is conservative.

The per-fold RMSE of the final ensemble components was:

Fold	Adap. LASSO RMSE	Stab. OLS RMSE
1	27.69	23.36
2	16.25	25.22
3	26.64	26.07
4	18.76	15.71
5	18.77	20.80
Mean $\pm$ std	21.6 $\pm$ 4.7	22.2 $\pm$ 3.9

The two models make complementary errors: Adaptive LASSO excels on Folds 2 and 5, while Stability OLS excels on Folds 1 and 4. Their weighted combination (0.60/0.40) achieves $\widehat{\text{RMSE}} = 21.18$ with $\hat{\sigma} = 3.70$, lower than either component alone.

Results

Our final predictor is a **weighted ensemble**:

$$\hat{y} = 0.60 \hat{y}^{\text{AdapLASSO}} + 0.40 \hat{y}^{\text{StabOLS}},$$

where both components use the same MICE pipeline (`BayesianRidge`, $t = 2$, `random.state=42`). The Adaptive LASSO uses Ridge-initialised weights with $\gamma = 1.0$. The Stability OLS component retains 8 features selected in more than 50% of $B = 200$ bootstrap sub-samples and refits OLS on this support.

Three features are always selected (frequency = 1.00): `x_32` (coefficient 56.7), `x_62` (34.3), `x_36` (27.5). Five additional features appear in 63%–81% of sub-samples. The remaining 107 features receive zero weight, confirming a sparse true signal.

The RMSE progression across methodological steps:

Step	Change	CV RMSE	$\hat{\sigma}$
Baseline	Median + ElasticNet	33.20	5.19
Step 1	KNN imputation ($k = 5$)	26.22	5.07
Step 2	MICE ($t = 2$)	23.20	2.86
Step 3	Adaptive LASSO ($\gamma = 1$)	22.07	4.69
Step 4	Stability Selection + OLS	22.15	3.92
Step 5	Ensemble (0.60/0.40)	21.18	3.70

Applied to the 1,000 test observations, the ensemble produces predictions in $[-14.2, 428.7]$ with mean 199.2 (training mean: 211.5).

Our estimate of the test RMSE is

$$\widehat{\text{RMSE}} = \mathbf{21.18}, \quad \hat{\sigma} = 3.70,$$

corresponding to an approximate 95% interval of $[17.5, 24.9]$. We report the conservative 5-fold CV score rather than a resubstitution error, in keeping with the principle that it is better to be honestly mediocre than confidently wrong.